Connected Subsets of $\mathbb{R}$ are Exactly Intervals

Conceptual Understanding

The idea of connectedness arises from the intuitive notion of a set being “in one piece”. A set is connected if it cannot be broken into two separate parts without creating a gap. In the real line $\mathbb{R}$, this intuitive idea becomes very precise because $\mathbb{R}$ has a natural order and a notion of points lying “between” other points.

If we imagine a subset of $\mathbb{R}$ drawn on a number line, then connectedness means that we can move from one point of the set to another without jumping over any missing points. If at any place a point is missing between two points of the set, then the set has a gap, and this gap allows us to split the set into two separate pieces.

> Thus, in $\mathbb{R}$, connectedness is equivalent to having no gaps.

Definition of Connected Set

A subset $A \subseteq \mathbb{R}$ is said to be connected if there do not exist two non-empty sets U and V such that:

$$A = U \cup V$$

and

$$U \cap V = \emptyset$$

where U and V are open in the subspace topology on A.

This means that A cannot be written as a union of two disjoint open subsets. If such a decomposition exists, then the set is said to be disconnected.

Definition of Interval

A subset $A \subseteq \mathbb{R}$ is called an interval if it contains every point lying between any two of its points.

Formally, A is an interval if for all $a, b \in A$ with $a < b$, and for every c satisfying:

$a < c < b$

we have:

$c \in A$

Thus, an interval has no missing interior points. It is “filled” completely between its elements.

Theorem

> A subset A ⊆ ℝ is connected if and only if A is an interval.

Proof

Part 1: If A is connected, then A is an interval

We prove this statement using contradiction. Suppose that A is connected, but A is not an interval.

Since A is not an interval, there must exist two points a and b in A such that a < b, and there exists some point c in ℝ satisfying:

$a < c < b$

but c does not belong to A. This means that there is a gap at the point c. The set includes points on both sides of c but does not include c itself.

This missing point allows us to divide A into two separate parts. Define the sets:

$U = A \cap (-\infty, c)$

$V = A \cap (c, \infty)$

The set U consists of all points of A that lie strictly to the left of c, and the set V consists of all points of A that lie strictly to the right of c.

We now show that U and V form a separation of A.

First, both (-∞, c) and (c, ∞) are open sets in ℝ. Therefore, their intersections with A are open in the subspace topology on A. Hence, U and V are open in A.

Second, U and V are disjoint because no point can be both less than c and greater than c at the same time.

Third, U ∪ V = A because every point of A is either less than c or greater than c, since c itself is not in A.

Fourth, both U and V are non-empty because a belongs to U and b belongs to V.

Thus, A is expressed as a union of two disjoint non-empty open sets, which means that A is disconnected.

This contradicts the assumption that A is connected.

Therefore, our assumption that A is not an interval must be false.

Hence, A is an interval.

Part 2: If A is an interval, then A is connected

We again use proof by contradiction. Suppose that A is an interval but is not connected.

Then there exist two disjoint non-empty open sets U and V in A such that:

$A = U \cup V$

Since U and V are both non-empty, we can choose points u ∈ U and v ∈ V. Without loss of generality, assume that:

$u < v$

Now consider the set U. Since U is bounded above by v, we can define:

$c = sup(U)$

The supremum c is the least upper bound of U. This means that:

- Every element of U is less than or equal to c
- There are points of U arbitrarily close to c from the left

Since u ≤ c ≤ v and A is an interval containing u and v, it must contain every point between them. Therefore:

$c \in A$

Now, since A = U ∪ V, the point c must belong either to U or to V.

Case 1: c ∈ U

Since U is open in A, there exists some ε > 0 such that:

$(c - \varepsilon, c + \varepsilon) \cap A \subseteq U$

This implies that there are points greater than c that also belong to U. But this contradicts the fact that c is an upper bound of U. Therefore, c cannot belong to U.

Case 2: c ∈ V

Since V is open in A, there exists some ε > 0 such that:

$(c - \varepsilon, c + \varepsilon) \cap A \subseteq V$

This implies that there are points less than c that belong to V. However, points less than c and sufficiently close to c must belong to U by the definition of supremum. This gives a contradiction.

Thus, c cannot belong to V either.

Since c must belong to either U or V but cannot belong to either, we obtain a contradiction.

Therefore, A cannot be disconnected.

Hence, A is connected.

Conclusion

> A subset of $\mathbb{R}$ is connected if and only if it is an interval.

Deep Explanation

This theorem works specifically in $\mathbb{R}$ because of two crucial properties.

First, $\mathbb{R}$ has an order structure, which allows us to talk about one point lying between two others. This idea of “betweenness” is essential in defining intervals.

Second, $\mathbb{R}$ satisfies the completeness property, meaning that every bounded set has a supremum. This property is used critically in the second part of the proof to construct the point c and derive a contradiction.

Without these properties, the theorem may fail in more general spaces.

Intuition Summary

If a set misses even a single point between two of its elements, that missing point creates a gap which allows the set to be split into two parts. This makes the set disconnected.

On the other hand, if a set has no such gaps, then any attempt to split it will fail, because boundary points will always lead to contradictions.

> Connected sets in $\mathbb{R}$ are precisely those sets that contain all points between their elements.